Identity-Compatible Auctions

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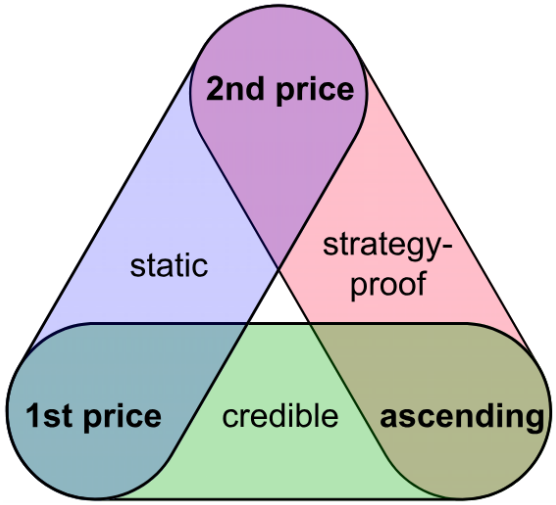
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Motivation

Common auction formats: 1st price (FPA), 2nd price (SPA), ascending, descending...

- Classic results: revenue equivalence
- But they differ a lot in implementation details of extensive forms.

An auction trilemma (Akbarpour and Li, 2020)



Motivation: One Person, Many Accounts

Online platforms observe accounts, not the people behind them.

Consider an eBay auction:

- The highest genuine bid is \$100; the second-highest is \$70.
- The seller secretly bids \$95 through another account.
- The genuine buyer still wins, but now pays \$95.

The seller gets an extra \$25.

Google also switched to FPAs in 2019 and conceals the number of bidders.

Q. Which formats remain robust when identities can be easily faked?

Shill Bidding and Identity Compatibility

Shill bidding: (i) The seller uses fake identities to pretend to be bidders, or (ii) a single buyer employs more than one identity to pose as multiple bidders.

Identity compatibility (ID-C): An agent has no incentive to fake identities.

"Players" in Auctions

- **Buyer (agent i):** a real agent $i \in B$ with purchase interest
  - Potential buyers $\mathbb{B} = \{1, 2, \dots\}$
  - A finite set of buyers B, drawn randomly from $\mathcal{B} = 2^{\mathbb{B}}$
  - Controls a set of bidders $N_i \ni i$
- **Bidder (identity):** an anonymous registered name with a random label
  - The only object the designer or any participant ever observes
  - Controlled by some buyer or by the seller
- **Seller (agent 0):** owner of the item, a player in the extensive form
  - May control a set S of shills (bidders inserted by the seller)
  - No transfers from own shills
- **Auctioneer (designer):** the marketplace with reputational concerns
  - Commits to a collection of games $((q^n, t^n))_{n \in \mathbb{N}}$
  - Discloses or conceals the count (lit vs. dark)

Focus on seller's shill bidding first.

Timing

Timing

1. The auctioneer commits to a collection of auctions $((q^n, t^n))_{n \in \mathbb{N}}$.
2. A set of buyers B is drawn but not revealed.
3. Buyers privately learn their types.
4. Each buyer and the seller simultaneously choose N_i and S.
5. The standard auction is played with bidder set

$$N = \left(\bigcup_{i \in B} N_i \right) \cup S$$

Standard Environment

Standard environment (symmetric IPV):

- n buyers in the set B, risk-neutral
- Buyer i has private value $\theta_i \overset{\text{i.i.d.}}{\sim} F$, supp $F \subseteq [0, 1]$.

Leading example in this talk: $F = U[0, 1]$

- Virtual value $v(\theta) = 2\theta - 1$
- Reserve price $\rho = 1/2$
- Equilibrium FPA bid

$$\beta^n(\theta) = \theta - \frac{\theta^n - 2^{-n}}{n\theta^{n-1}},$$

  i.e., $\mathbb{E}\left[\max\{\text{highest rival type}, \rho\} \mid \text{win}\right]$

Incentives for Shill Bidding

Incentives for Shill Bidding.

As a leading example, let buyer types be $(0.8, 0.3)$. The winner 0.8 should bid

$$\beta^1(0.8) = \frac{1}{2} = 0.5, \quad \beta^2(0.8) = \frac{89}{160} \approx 0.556, \quad \beta^3(0.8) = \frac{383}{640} \approx 0.598\ldots$$

and

$$\beta^n(\theta) \to 0.8 \quad (n \to \infty).$$

Faking a crowd raises the bid!

Q. Can the seller manipulate the winner's payment in common auction formats?

SPA: Posing as a Loser to Push Up the Price

The winner pays the second-highest bid: $\max\{0.3, \rho\} = 0.5$.

Ex post: easy

- Insert one shill bidding $0.8 - \varepsilon$
- Payment: $0.5 \to 0.8$

Ex ante: nothing (Izmalkov, 2004)

- Shill bid: an extra reserve, already optimal at $1/2$

SPA satisfies Bayesian seller ID-C, but fails ex-post seller ID-C

(Lit) FPA: Faking Competition

The winner pays its own bid:

$$\beta^2(0.8) \approx 0.556.$$

Ex ante, the seller can insert one **zero-bid shill**:

- Now $n = 3$, so "the room is more crowded."
- The winner's bid becomes $\beta^3(0.8)$:

$$0.556 \to 0.598.$$

Moreover,

$$\beta^\infty(0.8) = 0.8,$$

full extraction!

(Lit) FPA fails Bayesian seller ID-C

Lit and Dark Auctions

Lit Auctions

The number of bidders $|N|$ is common knowledge.

Dark Auctions

The number of bidders $|N|$ is concealed.

- No point for SPA
- Lit FPA vs dark FPA

Assumption: buyers hold symmetric priors over $|N| = |B|$,

$$p(n) = \Pr\left[|B| = n \mid i \in B\right]$$

Impossibility in Lit Auctions

Theorem 2 (Impossibility in Lit Auctions)

No optimal lit auction is ex-post seller ID-C.

Proof sketch.

1. Optimality pins the bill.
2. The FPA is the best-defended format.
3. Even the FPA fails.

Bayesian Seller ID-C

Definition 11

A lit auction (q, t) *is* **Bayesian seller identity-compatible** *if, for all* $|S| \in \mathbb{N}$*, and all* $\theta_S^1, \theta_S^2, \dots \in \Theta^{|S|}$*,*

$$\underbrace{\mathbb{E}_{B \in \mathcal{B}}}_{\text{over who shows up}} \left[\underbrace{\mathbb{E}_{\theta_B \in \Theta^{|B|}}}_{\text{over their values}} \left[\underbrace{\sum_{i \in B} t_i(\theta_B)}_{\substack{\text{revenue: real buyers' payments only} \\ \text{(a shill paying the seller is money} \\ \text{from his left pocket to his right)}}} \right] \right]$$

$$\geq \mathbb{E}_{B \in \mathcal{B}} \left[\mathbb{E}_{\theta_B \in \Theta^{|B|}} \left[\sum_{i \in B} t_i \left(\theta_B, \underbrace{\theta_S^{|B|}}_{\substack{\text{shill bids, planned ex ante:} \\ \text{may react to the disclosed} \\ \text{count } |B|, \text{ never to types}}} \right) \right] \right].$$

Ex-Post Seller ID-C

Definition 13

A lit auction (q, t) *is* **ex-post seller identity-compatible** *if, for all* $|S| \in \mathbb{N}$,

$$\underbrace{\mathbb{E}_{B \in \mathcal{B}}\left[\mathbb{E}_{\theta_B \in \Theta^{|B|}}\left[\sum_{i \in B} t_i(\theta_B)\right]\right]}_{\text{honest revenue, as before}}$$

$$\geq \mathbb{E}_{B \in \mathcal{B}}\left[\mathbb{E}_{\theta_B \in \Theta^{|B|}}\left[\underbrace{\sup_{\theta_S \in \Theta^{|S|}}}_{\substack{\text{bids now chosen ex post:} \\ \text{types drawn and seen,} \\ \text{best response realization} \\ \text{by realization}}} \sum_{i \in B} t_i(\theta_B, \theta_S)\right]\right].$$

A Stronger Condition: Auctioneer ID-C

Q. Among all lit auctions that are ex-post seller ID-C, which maximizes expected revenue?

A condition stronger than ex-post seller ID-C: ex-post auctioneer ID-C

1. **No commitment to $|S|$:** the auctioneer may leak the number of buyers, or the seller may be the auctioneer.
2. **Untrustworthy randomization device:**
   - With rule "sell to i with prob. q_i," the auctioneer always sells.
   - Buyer i effectively faces the conditional price t_i/q_i.
   - Ties are broken toward whoever pays most conditional on winning.

Untrustworthy Randomization

Example *If $\theta_i \geq 0.6$, sell to i with prob. $1/2$ at price 0.6; otherwise no sale*

$$q_i = \tfrac{1}{2}, \qquad t_i = 0.3, \qquad t_i/q_i = 0.6$$

Run honestly $\Longrightarrow$ revenue 0.3; skip the coin flip and always sell $\Longrightarrow$ revenue 0.6

- The price was never altered.
- Only the randomization is rigged.
- Buyer i observes the reasonable outcome "I won and paid 0.6".

So t_i/q_i is the most that can be extracted.

Ex-Post Auctioneer ID-C

Definition 14

A lit auction (q, t) *is* **ex-post auctioneer identity-compatible** *if*

$$\underbrace{\mathbb{E}_{B\in\mathcal{B}}\left[\mathbb{E}_{\theta_B\in\Theta^{|B|}}\left[\sum_{i\in B} t_i(\theta_B)\right]\right]}_{\text{honest revenue, as before}}$$

$$\geq \mathbb{E}_{B\in\mathcal{B}}\left[\mathbb{E}_{\theta_B\in\Theta^{|B|}}\left[\sup_{\substack{i\in B,|S|\in\mathbb{N}\cup\{0\}\\ \theta_S\in\Theta^{|S|},q_i(\theta_B,\theta_S)>0}} \underbrace{\frac{t_i(\theta_B,\theta_S)}{q_i(\theta_B,\theta_S)}}_{\substack{\text{price conditional on winning:}\\ \text{the rigged device always sells,}\\ \text{so } q_i \text{ is never wasted}}}\right]\right].$$

The Posted-Price Mechanism

The posted-price mechanism

- The seller announces a price c.
- Each bidder "accepts" or "rejects".
- One accepting bidder is drawn uniformly and pays c.
- Otherwise, no sale.

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Theorem 3 (Tying the Seller's Hands)

The posted-price mechanism maximizes expected revenue among all lit auctions that are ex-post auctioneer ID-C.

Posted Price Is Not a Reserve Price

Remark posted price $\neq$ reserve price

- Suppose $p(1) = p(2) = 1/2$.
- A buyer accepts if and only if $\theta_i \geq c$:

$$\max_c \left\{ \frac{1}{2} c(1-c) + \frac{1}{2} c(1-c^2) \right\}.$$

The FOC gives

$$c = \frac{\sqrt{7}-1}{3} \neq \frac{1}{2}.$$

- A reserve only screens out low types; the posted price also absorbs the competition from a varying headcount.

Dark FPA

Buyers cannot see n in the dark FPA. Start by solving for the bid.

Let Y be the highest rival type. Its cdf

$$G(\theta) := \Pr[Y < \theta]$$

is the probability of winning. Then

$$\beta(\theta) = \mathbb{E}[\max\{Y, \rho^*\} | Y < \theta] = \theta - \frac{\int_{\rho^*}^{\theta} G(x)dx}{G(\theta)}.$$

The only difference is the rival distribution:

	Highest-rival CDF G
Lit: count known	$G = F^{n-1}$
Dark: count hidden	$G = \sum_n p(n)F^{n-1}$

Why Shill Bidding Fails in a Dark FPA

Shill bidding fails even with full information:

- Shill bids $< \beta(0.8)$: the winner is unchanged and pays his unchanged bid.
- Shill bids $> \beta(0.8)$: the shill wins and generates zero revenue.

Pay-as-bid
Price equals the winner's own bid, independent of losers.

Hidden headcount
A shill cannot change the "perceived count".

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Theorem 4 (Light in the Dark)

The dark FPA with the reserve price ρ^* is the unique optimal dark auction that is ex-post auctioneer ID-C and Bayesian buyer ID-C.

Buyer ID-C

Now we proceed to consider ID-C on the buyer side.

	Seller	Buyer
Goal	Push the price up	Pull the price down
Weapon	Shills fake a crowd	Clones grab ties or expand the bid space

SPA Blocks Clones

SPA blocks clones. The price for the winner is the highest *other* bid.

- Possible clones:

$$\text{a clone bid} \in \left(\frac{1}{2}, 0.8\right] \implies \text{your own price rises}$$

$$\text{a clone bid} > 0.8 \implies \text{a clone wins and overpays}$$

$$\text{a clone bid} < \frac{1}{2} \implies \text{nothing}$$

SPA satisfies ex-post buyer ID-C

Strategy-Proofness Is Not Enough

Strategy-Proofness is not enough. The posted price is strategy-proof.

- Yet ties are split among all acceptors.

Clones of buyer 1	Acceptors	Buyer 1 winning probability
1	2	$\frac{1}{2}$
2	3	$\frac{2}{3}$
k	$k+1$	$\frac{k}{k+1} \to 1$

Bayesian Buyer ID-C

Definition

A lit auction (q, t) *is* **Bayesian buyer identity-compatible** *if, for all* $i \in \mathbb{B}$*, all* $\theta_i \in \Theta$*, all* $|N_i| \in \mathbb{N}$*, and all* $\hat{\theta}^1_{N_i}, \hat{\theta}^2_{N_i}, \cdots \in \Theta^{|N_i|}$*,*

$$\underbrace{\mathbb{E}_{B \in \mathcal{B}}\left[\mathbb{E}_{\theta_{-i} \in \Theta^{|B|-1}}\left[\theta_i q_i(\theta_i, \theta_{-i}) - t_i(\theta_i, \theta_{-i})\right] \,\middle|\, \underbrace{i \in B}_{\substack{\text{given } i \text{ is} \\ \text{in the market}}}\right]}_{\substack{\text{buyer } i\text{'s expected payoff with a single identity, bidding truthfully}}}$$

$$\geq \mathbb{E}_{B \in \mathcal{B}}\left[\mathbb{E}_{\theta_{-i} \in \Theta^{|B|-1}}\left[\underbrace{\sum_{j \in N_i}\left[\theta_i q_j\left(\overbrace{\hat{\theta}^{|B|}_{N_i}}^{\substack{\text{clone-bid profile, planned} \\ \text{ex ante: may react to the} \\ \text{count, never to types}}}, \theta_{-i}\right) - t_j\left(\hat{\theta}^{|B|}_{N_i}, \theta_{-i}\right)\right]}_{\substack{\text{payoff summed over all his clones:} \\ \text{any clone's win is his win, every clone's bill is his bill}}}\right] \,\middle|\, i \in B\right].$$

Ex-Post Buyer ID-C

Definition 10

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Ex-Post Buyer ID-C and Strategy-Proofness

Lemma 1

Ex-post buyer ID-C implies strategy-proofness.

Proof. Letting $|N_i| = 1$ in the definition yields

$$\theta_i q_i - t_i = \sup_{\hat{\theta}_i} \left[\theta_i q_i(\hat{\theta}_i, \theta_{-i}) - t_i(\hat{\theta}_i, \theta_{-i}) \right].$$

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Theorem 1

The SPA with the reserve price ρ^* is the unique optimal auction that is ex-post buyer ID-C.

FPA and Buyer ID-C

Ex-post buyer ID-C fails in both FPAs, since strategy-proofness fails.

Lit FPA

- Pay-as-bid: clones cannot lower his payment.
- Clones raise the disclosed count, so rivals bid $\beta^{n+|N_i|} > \beta^n$.
- The very force that the seller exploits harms the buyer.

Dark FPA

- The headcount is invisible.
- Rivals do not react.
- Clones are pure nothing.

Both FPAs satisfy Bayesian buyer ID-C, but fail ex-post buyer ID-C

Summary of Identity Compatibility

	Seller ID-C		Buyer ID-C			
	Bayesian	Ex post	Bayesian	Ex post	Lit	Optimal
SPA	✓	✗	✓	✓	✓	✓
Lit FPA	✗	✗	✓	✗	✓	✓
Dark FPA	✓	✓	✓	✗	✗	✓
Posted Price	✓	✓	✗	✗	✓	✗

A Dilemma: ID-C for Both Parties

The two sides' demands directly conflict:

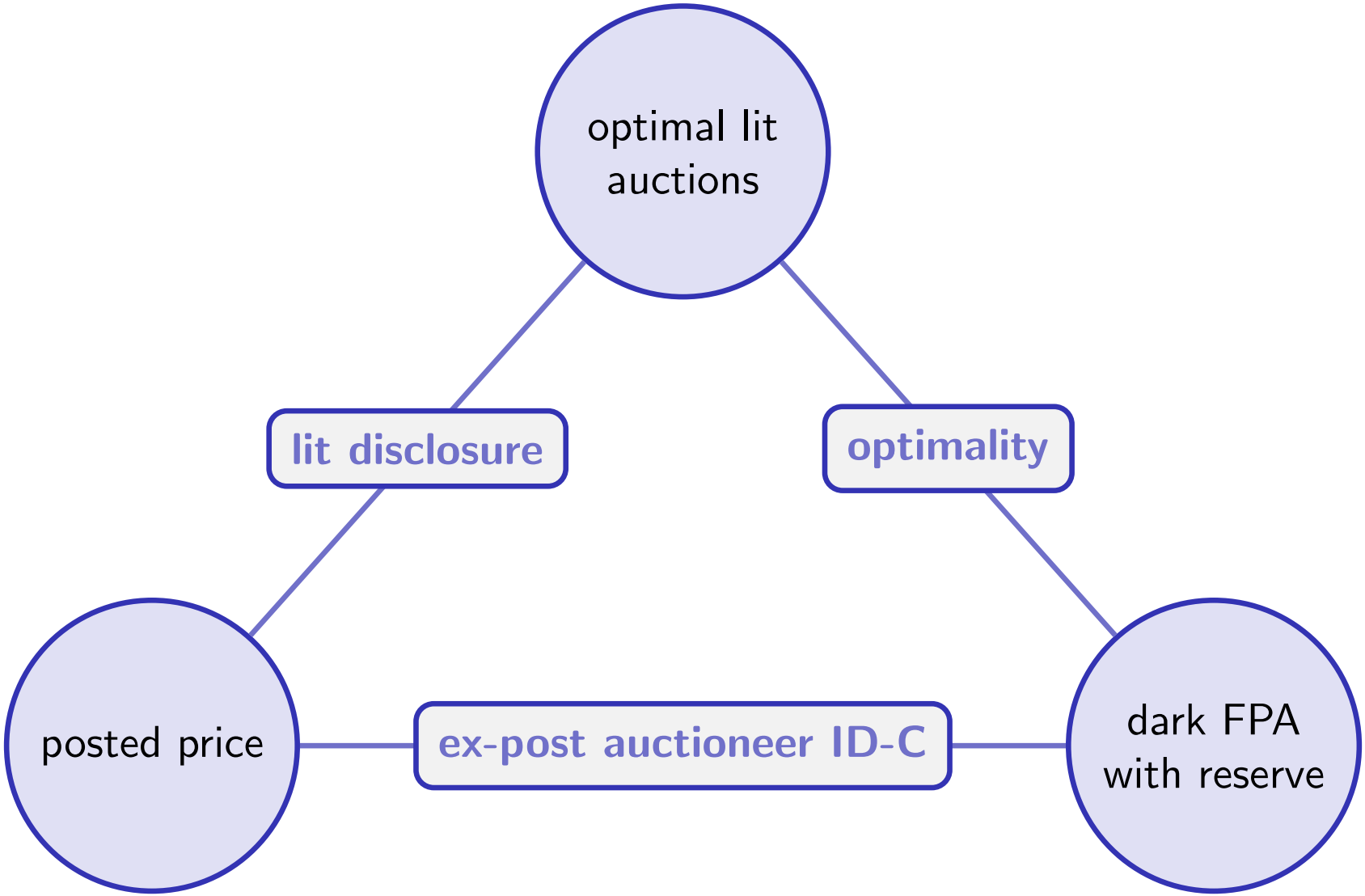
- To block the buyer from clone bidding, the price should not depend on the winner's type $\Longrightarrow$ SPA-type
- To block the seller from shill bidding, the price should not depend on the losers' types $\Longrightarrow$ FPA-types

So one must relax one side to Bayesian:

- SPA: blocks the buyer fully (ex-post) + holds the seller (Bayesian)
- Dark FPA: holds the buyer (Bayesian) + blocks the seller fully (ex-post)

Which one to pick depends on which side is better informed.

Another Trilemma



Disclosure Policy

Q. Is lit versus dark all-or-nothing?

No. The designer can partially reveal the count.

- Example: "run the SPA with at most 5 bidders, and the dark FPA otherwise."
- A binary signal: $\leq$ or $>$ 5

Disclosure Policy

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- A binary signal: $\leq$ or > 5

Disclosure Policy

A **disclosure policy** ϕ *is a surjective mapping from the number of agents* $\mathbb{N}$ *to a set of signals* Y*. It is* **partitional** *if, for any* $y, y' \in Y$*,*

$$\phi^{-1}(y) \cap \phi^{-1}(y') \neq \varnothing \quad \Longrightarrow \quad y = y'.$$

Partitional Disclosure

A partitional policy is a partition of $\mathbb{N}$: buyers learn which cell the count falls in.

- Example: $Y = \{\text{small}, \text{large}\}$

$$\phi(n) = \begin{cases} \text{small}, & n \leq 5, \\ \text{large}, & n \geq 6 \end{cases}$$

- Lit and dark are the two extremes:

$$\underbrace{Y^{\text{lit}} = \mathbb{N},\ \phi^{\text{lit}} = \text{id}_{\mathbb{N}}}_{\text{finest}} \qquad \underbrace{Y^{\text{dark}} = \{d\},\ \phi^{\text{dark}} \equiv d}_{\text{coarsest}}$$

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Theorem 6

Any partitional mechanism induces an equivalent dark mechanism.

Beliefs

The honest profile is *an* equilibrium. No uniqueness.

- ID-C is the condition that "no shills or suspicion" is self-enforcing.
- Similar to Akbarpour and Li (2020):

  credible + BIC $\iff$ the rule-following profile is a BNE of the messaging game

Suspicion equilibria may be consistent.

- "Shills with prob. p" is a possible belief when paired with actual shilling.
- Nothing eliminates it; related work: Chakraborty and Kosmopoulou (2004), Lamy (2009)

Implementation via Dynamic Formats

Much harder to deal with!

Two-stage auction (Seibe and Skoda, 2023). There are K slots.

1. Preselection: the K highest bidders enter the final stage.
2. Final stage: the selected bidders compete in an ascending auction.

With K identities under one buyer's control,

$$\underbrace{\text{submit } K \text{ high bids}}_{\text{occupy every slot}} \Longrightarrow \underbrace{\text{withdraw } K-1 \text{ identities}}_{\text{leave one identity alone}} \Longrightarrow \text{win near the reserve}$$

Takeaways

This paper studies **identity compatibility** in auctions and formalizes the concepts of **lit** and **dark** auctions.

	Seller ID-C		Buyer ID-C			
	Bayesian	Ex post	Bayesian	Ex post	Lit	Optimal
SPA	✓	✗	✓	✓	✓	✓
Lit FPA	✗	✗	✓	✗	✓	✓
Dark FPA	✓	✓	✓	✗	✗	✓
Posted Price	✓	✓	✗	✗	✓	✗

- **A dilemma:** no optimal auction is both ex-post buyer and ex-post seller ID-C. One side must be relaxed to Bayesian.
- **Another trilemma:** (i) optimality, (ii) ex-post auctioneer ID-C, and (iii) lit disclosure